DEEP PUNJ, PhD

Data Scientist and Data Analyst | Python, SQL, Power BI and Machine Learning | PhD in Optics

Python · pandas · scikit-learn · SQL · PostgreSQL · Power BI · data quality · forecasting · lasers · OCT · imaging data

Ghent, Belgium | deep.punj@outlook.com | +32 472 72 29 07 | punjdeep.com | linkedin.com/in/deeppunj | github.com/deeppunj

PROFILE

PhD physicist with more than ten years of analyzing complex measurement datasets. At Ghent University (Faculty of Pharmaceutical Sciences) I write Python programs for image analysis, instrument control and routine quality checks of laser systems. My personal projects on GitHub cover a reporting pipeline with data-quality rules, a PostgreSQL star schema and a Power BI report; machine learning on thermal-camera images and on synthetic health records; a price forecast feeding a battery schedule; and a photonics simulation. I also write the LinkedIn teaching series Light meets Data. I explain results clearly to students, colleagues and non-specialists, and have worked in teams in six countries. I enjoy working out what data is really saying.

CORE SKILLS

- **BI and visualization:** Power BI data model on a star schema (relationships set by hand, single-direction filters, marked date table), DAX measures, data types in Power Query; Streamlit dashboards, Tableau, Excel.
- **Data modeling and pipelines:** Kimball star schema (fact and dimension tables) in PostgreSQL with primary and foreign keys; Python ETL pipeline from raw CSV files to clean tables and database load; KPI definitions.
- **SQL:** PostgreSQL; joins, CTEs, window functions (shares of totals, 7-day moving averages) and aggregations for KPI reports.
- **Data quality:** in the fab reporting project, rules that catch duplicates, missing or unknown tool IDs and impossible timestamps, and write each case to a data-quality log; checks that every fact row links to a dimension row; dashboard figures checked once against the SQL results. In the patient-finding project, three pytest tests against data leakage; in the laser project, tests run by GitHub Actions on every push.
- **Python for data analysis:** pandas, NumPy, SciPy, Matplotlib; scikit-learn (Random Forest, logistic regression, cross-validation, baselines); MLflow; R (basic).
- **Statistics:** descriptive statistics, weekly KPI trends, breakdowns by tool group and hold reason.
- **Time series and forecasting:** price forecasting split by time (train, validate, test years); constrained optimization with cvxpy.
- **Documentation:** KPI dictionary, data model description, test plan and README guides for each project.
- **Data platforms:** Databricks course in progress (PySpark, Delta Lake; Bronze, Silver and Gold lakehouse tables built in the course workspace).
- **Tools:** Git and GitHub, PyCharm, VS Code, Jupyter, conda; LLM tools (Claude, NotebookLM). Languages: English (fluent), Hindi (native).

PERSONAL PROJECTS (GITHUB)

Pilot-line production reporting | personal project, simulated data | github.com/deeppunj/fab-reporting-ba

- Python ETL pipeline turns raw lot, tool and yield records (609 lots, 8 tool groups) into a Kimball star schema loaded into PostgreSQL with enforced foreign keys. Data-quality rules log 89 faulty move records: 40 duplicates removed, 34 missing or unknown tool IDs mapped to an UNKNOWN tool, and 15 track-outs before track-in kept but left out of time averages. Three checks between fact and dimension tables return zero orphans.
- SQL KPI queries with CTEs and window functions: on-time delivery 72.9%, weekly trends, utilization and queue time by tool group; they locate the bottleneck built into the simulation (lithography, 45% of all queue time).
- Power BI report on the star schema (DAX measures for on-time %, utilization and queue share) and a Streamlit dashboard; the repo documents each KPI (definition, formula, owner, target), the data model and a test plan in which six headline numbers in the dashboard match the SQL results.

Patient finding | personal project, synthetic data (Synthea) | github.com/deeppunj/patient-finding-synthea

- SQL cohort and features in PostgreSQL, three pytest tests against data leakage, MLflow tracking; the top 10% of patients by risk score hold 49% of next year's kidney-disease cases (synthetic data only).

Laser weld defect inspection | personal project, public lab data | github.com/deeppunj/laser-defect-inspection-ml

- Random Forest on 7 thermal features from 16,362 camera frames of 97 real laser welds, cross-validation split by weld; ROC AUC 0.974 ± 0.007, balanced accuracy 0.927; 31 automated tests, run by GitHub Actions on every push.

Battery dispatch, Belgium | personal project, historical price data | github.com/deeppunj/battery-dispatch-be

- Gradient-boosting forecast of Belgian day-ahead prices (train 2015 to 2022, validate 2023, test 2024): mean absolute error 17.2 against 23.2 EUR/MWh for a same-hour-yesterday forecast; feeds a cvxpy charging schedule.

SiN edge-coupler study | personal project, simulation only | github.com/deeppunj/sipho-coupler-study

- Python mode solver for a silicon-nitride edge coupler at 1550 nm, checked against the exact slab solution; best modelled fibre coupling loss 0.43 dB at a 250 nm tip; Monte Carlo assembly yield at a 1 dB spec of 96% at 0.25 µm placement accuracy against 19% at 1 µm.

PROFESSIONAL EXPERIENCE

Optical Engineer / Researcher | Ghent University, Faculty of Pharmaceutical Sciences, Belgium *Apr 2022 – present*

- Write Python programs for image analysis, instrument control (cameras, stages) and routine quality checks of the laser systems.
- Use Claude and NotebookLM for data analysis support, report drafting and SOP preparation, and check the results against the measured data.
- Designed and built a retinal photoporation setup that combines an OCT system with nanosecond lasers at 532 and 1064 nm, with the laser spot co-aligned to the OCT image; installed in the UZ Gent Animalarium core facility.
- Rebuilt and extended a nanosecond OPO platform; check its performance across wavelengths on a schedule and fix faults.
- Co-managed the move of the whole lab to a new location against a deadline and recommissioned four microscopy setups; handled equipment purchases, including a nanosecond laser.
- Train PhD students and postdocs, teach in the annual Specialist Microscopy course, and act as local IT administrator (lab PCs, group website, SharePoint).
- Contribute to funding proposals and research papers.

Data Mining Intern | World Development Organization *Nov 2021*

- Wrote Python scrapers (requests, BeautifulSoup) to collect NGO funding data from several websites, then cleaned and merged it with pandas into one dataset.

Postdoctoral Researcher, Single-Molecule Optics | Leiden University, Netherlands (Orrit's group) *Jun 2019 – Apr 2022*

- Made gold nanorod dimers by wet-chemical synthesis, did SEM measurement and analysis and COMSOL simulations, and measured and analyzed one- and two-photon fluorescence enhancement of single molecules.
- Mentored a PhD student and supervised a B.Sc. thesis project.

Postdoctoral Researcher | IIT Madras · UC Irvine · KAUST *2015 – 2019*

- KAUST (Jun 2015 – Jun 2016): FIB nanofabrication on Quanta 3D and Helios systems, and design and fabrication of AFM probe shapes for near-field measurements.
- UC Irvine (Apr 2017 – Mar 2018): fabricated plasmonic nanostructures with a nanorobot inside the SEM and modelled gold nanoparticle assemblies in Lumerical FDTD.
- IIT Madras (Aug 2018 – Mar 2019): plasmonics-assisted ptychography for high-resolution imaging.

PhD Researcher, Nano-Optics | Aix-Marseille Université, Institut Fresnel (CNRS), France *Mar 2012 – Feb 2015*

- Modelled gold nanoantennas with FDTD and measured them with fluorescence correlation spectroscopy.

EDUCATION

PhD, Nano-Optics | Aix-Marseille Université, Institut Fresnel, France *2012 – 2015*

Integrated M.Sc., Photonics | Cochin University of Science and Technology, India *2006 – 2011*

TRAINING AND CERTIFICATIONS

- Databricks Data Engineer Associate: Practical Guide (Packt on Coursera, in progress): PySpark, Delta Lake
- Google Data Analytics Professional Certificate (SQL, R, Tableau, data cleaning)
- IBM: Python for Data Science, AI and Development; Python Project for Data Science; Data Science Foundations Level 1; Data Visualization with Excel and Cognos
- Breast Cancer Prediction Using Machine Learning (Coursera Project Network); Tableau Essential Training and MATLAB 2018 Essential Training (LinkedIn Learning)

PUBLICATIONS

20 journal papers, 4 as first author; h-index 13, about 1,100 citations (Google Scholar). Recent co-authored work in Science Advances and Advanced Science (2026).